VERIFYING THE AXIOMS AFTER A FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file weil.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the linear algebra principles transferring the formal axioms, the compatibility of the cycle class map with base change through Chern characters and the splitting principle, and the treatment of axioms (A6) and (A9) through the Kunneth decomposition of the diagonal and the uniqueness of the degree map.

Editorial note. The label lemma-check-over-extension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source weil.tex –label lemma-check-over-extension –start 5444 –end 5616 –local-id 28125 returned RESULT=OK (L28125 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The cycle class map as the Chern character of a K theory class. Same chapter, the construction of the cycle class map from the Chern character using axioms (A1) to (A4). Expresses the cycle class map as the Chern character so that compatibility with base change can be checked on Chern classes of vector bundles.
- Compatibility of the base change map on Chow groups with capping by Chern classes and the splitting principle. Chow Homology, the lemma on the pullback along a base change of the base scheme, its compatibility with Chern classes, and the splitting principle for vector bundles on projective space bundles. Reduces the commutation of the cycle class map with base change to the agreement of first Chern classes, which is one of the hypotheses.
- Uniqueness of the degree map determined by axiom (A6). Same chapter, the remark on the trace map, and the construction in the proof of axiom (A6). Identifies the degree maps after base change, which is what makes axiom (A9) follow by a diagram chase.

Source reference. Original lemma 0FVJ: sources/stacks-chow/tags.txt, source lines 19814–19814. Prerequisite pointer invoked at proof line(s) 5528; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0FVN: sources/stacks-chow/tags.txt, source lines 19818–19818. Prerequisite pointer invoked at proof line(s) 5535; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02UL: sources/stacks-chow/tags.txt, source lines 3736–3736. Prerequisite pointer invoked at proof line(s) 5548; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0FVE: sources/stacks-chow/tags.txt, source lines 19809–19809. Prerequisite pointer invoked at proof line(s) 5549; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0FVF: sources/stacks-chow/tags.txt, source lines 19810–19810. Prerequisite pointer invoked at proof line(s) 5525; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FIF: sources/stacks-weil/pinned-original-source.tex, source lines 4713–4713. Prerequisite pointer invoked at proof line(s) 5523; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0FIK: sources/stacks-weil/pinned-original-source.tex, source lines 4929–4929. Prerequisite pointer invoked at proof line(s) 5606; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k'/k be an extension of fields. Let F'/F be an extension of fields of characteristic 0. Assume given*

(1) *data (D0), (D1), (D2') for k and F denoted $F(1), H^*, c_1^H$,*
(2) *data (D0), (D1), (D2') for k' and F' denoted $F'(1), (H')^*, c_1^{H'}$, and*
(3) *an isomorphism $F(1) \otimes_F F' \to F'(1)$, functorial isomorphisms $H^*(X) \otimes_F F' \to (H')^*(X_{k'})$ on the category of smooth projective schemes X over k such that the diagrams*

$$\begin{array}{ccc} \mathrm{Pic}(X) & \xrightarrow{c_1^H} & H^2(X)(1) \\ \downarrow & & \downarrow \\ \mathrm{Pic}(X_{k'}) & \xrightarrow{c_1^{H'}} & (H')^2(X_{k'})(1) \end{array}$$

commute.

In this case, if $F'(1), (H')^*, c_1^{H'}$ satisfy axioms (A1) – (A9), then the same is true for $F(1), H^*, c_1^H$.

Proof. We go by the axioms one by one.

Axiom (A1). We have to show $H^*(\emptyset) = 0$ and that $(i^*, j^*) : H^*(X \amalg Y) \to H^*(X) \times H^*(Y)$ is an isomorphism where i and j are the coprojections. By the functorial nature of the isomorphisms $H^*(X) \otimes_F F' \to (H')^*(X_{k'})$ this follows from linear algebra: if $\varphi : V \to W$ is an F-linear map of F-vector spaces, then φ is an isomorphism if and only if $\varphi_{F'} : V \otimes_F F' \to W \otimes_F F'$ is an isomorphism.

Axiom (A2). This means that given a morphism $f : X \to Y$ of smooth projective schemes over k and an invertible $\mathcal{O}_Y$-module $\mathcal{N}$ we have $f^*c_1^H(\mathcal{L}) = c_1^H(f^*\mathcal{L})$. This is immediately clear from the corresponding property for $c_1^{H'}$, the commutative diagrams in the lemma, and the fact that the canonical map $V \to V \otimes_F F'$ is injective for any F-vector space V.

Axiom (A3). This follows from the principle stated in the proof of axiom (A1) and compatibility of c_1^H and $c_1^{H'}$.

Axiom (A4). Let $i : Y \to X$ be the inclusion of an effective Cartier divisor over k with both X and Y smooth and projective over k. For $a \in H^*(X)$ with $i^*a = 0$ we have to show $a \cup c_1^H(\mathcal{O}_X(Y)) = 0$. Denote $a' \in (H')^*(X_{k'})$ the image of a. The assumption implies that $(i')^*a' = 0$ where $i' : Y_{k'} \to X_{k'}$ is the base change of i. Hence we get $a' \cup c_1^{H'}(\mathcal{O}_{X_{k'}}(Y_{k'})) = 0$ by the axiom for $(H')^*$. Since $a' \cup c_1^{H'}(\mathcal{O}_{X_{k'}}(Y_{k'}))$ is the image of $a \cup c_1^H(\mathcal{O}_X(Y))$ we conclude by the princple stated in the proof of axiom (A2).

Axiom (A5). This means that $H^*(\mathrm{Spec}(k)) = F$ and that for X and Y smooth projective over k the map $H^*(X) \otimes_F H^*(Y) \to H^*(X \times Y)$, $a \otimes b \mapsto p^*(a) \cup q^*(b)$ is an isomorphism where p and q are the projections. This follows from the principle stated in the proof of axiom (A1).

We interrupt the flow of the arguments to show that for every smooth projective scheme X over k the diagram

$$\begin{array}{ccc} \mathrm{CH}^*(X) & \xrightarrow{\gamma} & \bigoplus H^{2i}(X)(i) \\ \downarrow{\scriptstyle g^*} & & \downarrow \\ \mathrm{CH}^*(X_{k'}) & \xrightarrow{\gamma'} & \bigoplus (H')^{2i}(X_{k'})(i) \end{array}$$

commutes. Observe that we have γ as we know axioms (A1) – (A4) already; see Lemma 0FIF. Also, the left vertical arrow is the one discussed in Chow Homology, Section 0FVF for the morphism of base schemes $g : \mathrm{Spec}(k') \to \mathrm{Spec}(k)$. More precisely, it is the map given in Chow Homology, Lemma 0FVJ. Pick $\alpha \in \mathrm{CH}^*(X)$. Write $\alpha = ch(\beta) \cap [X]$ in $\mathrm{CH}^*(X) \otimes \mathbf{Q}$ for some $\beta \in K_0(\mathit{Vect}(X)) \otimes \mathbf{Q}$ so that $\gamma(\alpha) = ch^H(\beta)$; this is our construction of γ. Since the map of Chow Homology, Lemma 0FVJ is compatible with capping with Chern classes by Chow Homology, Lemma 0FVN we see that $g^*\alpha = ch((X_{k'} \to X)^*\beta) \cap [X_{k'}]$. Hence $\gamma'(g^*\alpha) = ch^{H'}((X_{k'} \to X)^*\beta)$. Thus commutativity of the diagram will hold if for any locally free $\mathcal{O}_X$-module $\mathcal{E}$ of rank r and $0 \leq i \leq r$ the element $c_i^H(\mathcal{E})$ of $H^{2i}(X)(i)$ maps to the element $c_i^{H'}(\mathcal{E}_{k'})$ in $(H')^{2i}(X_{k'})(i)$. Because we have the projective space bundle formula for both X and X' we may replace X by a projective space bundle over X finitely many times to show this. Thus we may assume $\mathcal{E}$ has a filtration whose graded pieces are invertible $\mathcal{O}_X$-modules $\mathcal{L}_1, \ldots, \mathcal{L}_r$. See Chow Homology, Lemma 02UL and Remark 0FVE. Then $c_i^H(\mathcal{E}$ is the ith elementary symmetric polynomial in $c_1^H(\mathcal{L}_1), \ldots, c_1^H(\mathcal{L}_r)$ and we conclude by our assumption that we have agreement for first Chern classes.

Axiom (A6). Suppose given F-vector spaces V, W, an element $v \in V$, and a tensor $\xi \in V \otimes_F W$. Denote $V' = V \otimes_F F'$, $W' = W \otimes_F F'$ and v', ξ' the images of v, ξ in V', $V' \otimes_{F'} W'$. The linear algebra principle we will use in the proof of axiom (A6) is the following: there exists an F-linear map $\lambda : W \to F$ such that $(1 \otimes \lambda)\xi = v$ if and only if there exists an F'-linear map $\lambda' : W \otimes_F F' \to F'$ such that $(1 \otimes \lambda')\xi' = v'$.

Let X be a nonempty equidimensional smooth projective scheme over k of dimension d. Denote $\gamma = \gamma([\Delta])$ in $H^{2d}(X \times X)(d)$ (unadorned fibre products will be over k).

Observe/recall that this makes sense as we know axioms (A1) – (A4) already; see Lemma 0FIF. We may decompose

$$\gamma = \sum \gamma_i, \quad \gamma_i \in H^i(X) \otimes_F H^{2d-i}(X)(d)$$

in the Künneth decomposition. Similarly, denote $\gamma' = \gamma([\Delta']) = \sum \gamma'_i$ in $(H')^{2d}(X_{k'} \times_{k'} X_{k'})(d)$. By the linear algebra princple mentioned above, it suffices to show that γ_0 maps to γ'_0 in $(H')^0(X) \otimes_{F'} (H')^{2d}(X')(d)$. By the compatibility of Künneth decompositions we see that it suffice to show that γ maps to γ' in

$$(H')^{2d}(X_{k'} \times_{k'} X_{k'})(d) = (H')^{2d}((X \times X)_{k'})(d)$$

Since $\Delta_{k'} = \Delta'$ this follows from the discussion above.

Axiom (A7). This follows from the linear algebra fact: a linear map $V \to W$ of F-vector spaces is injective if and only if $V \otimes_F F' \to W \otimes_F F'$ is injective.

Axiom (A8). Follows from the linear algebra fact used in the proof of axiom (A1).

Axiom (A9). Let X be a nonempty smooth projective scheme over k equidimensional of dimension d. Let $i : Y \to X$ be a nonempty effective Cartier divisor smooth over k. Let λ_Y and λ_X be as in axiom (A6) for X and Y. We have to show: for $a \in H^{2d-2}(X)(d-1)$ we have $\lambda_Y(i^*(a)) = \lambda_X(a \cup c_1^H(\mathcal{O}_X(Y))$. By Remark 0FIK we know that $\lambda_X : H^{2d}(X)(d) \to F$ and $\lambda_Y : H^{2d-2}(Y)(d-1)$ are uniquely determined by the requirement in axiom (A6). Having said this, it follows from our proof of axiom (A6) for H^* above that $\lambda_X \otimes \mathrm{id}_{F'}$ corresponds to $\lambda_{X_{k'}}$ via the given identification $H^{2d}(X)(d) \otimes_F F' = H^{2d}(X_{k'})(d)$. Thus the fact that we know axiom (A9) for $F'(1), (H')^*, c_1^{H'}$ implies the axiom for $F(1), H^*, c_1^H$ by a diagram chase. This completes the proof of the lemma. □